

Description

This application note aims to provide an overview of the topology discovery mechanism supported by the 10BASE-T1S technology. The limitations and system requirements for this application will be shown. Upon completion, the reader will understand the basic workings of topology discovery, how to start the topology discovery process, what the nodes on the network do during this process, and how to read the results given by the measuring device.

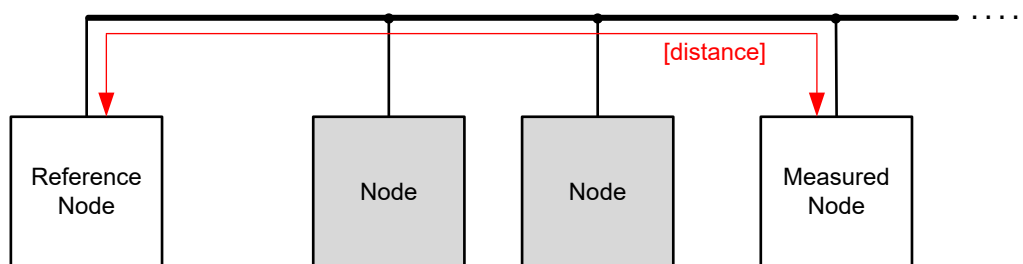


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1. References

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- *10BASE-T1S PMD Transceiver Interface Specification*, Version 1.5, OPEN Alliance, 2022.
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2. Introduction

The topology discovery process is used to measure the time required for a signal to travel between two 10BASE-T1S nodes on a network mixing segment. The physical distance between nodes is derived from a time-of-flight measurement on the physical medium transmission line. The process to make the necessary measurements and perform the calculation is defined in the OPEN Alliance *10BASE-T1S Topology Discovery* specification.

The topology discovery process requires the coordination of all nodes on the mixing segment. During the process, two nodes will have a designated function as either the *reference* or *measured* node. During the process, none of the other nodes on the mixing segment may transmit any signal, pulse, or message on the line. This application note details the role of each node and how to implement the measurement using standard registers. It does not suggest a particular management protocol.

3. Theory of Operation

System Configuration

The topology discovery process includes multiple measurement steps, each of which require the participation of the entire mixing segment. One node is designated as the reference node and a separate node is designated as the measured node. During measurement, only these two nodes will transmit signals, which are special signals used only during topology discovery. No other signals, including beacons, may be transmitted by any node on the segment during this time. Each node of the mixing segment network must be configured into its desired mode using an appropriate management protocol. Details of such a protocol are beyond the scope of this document.

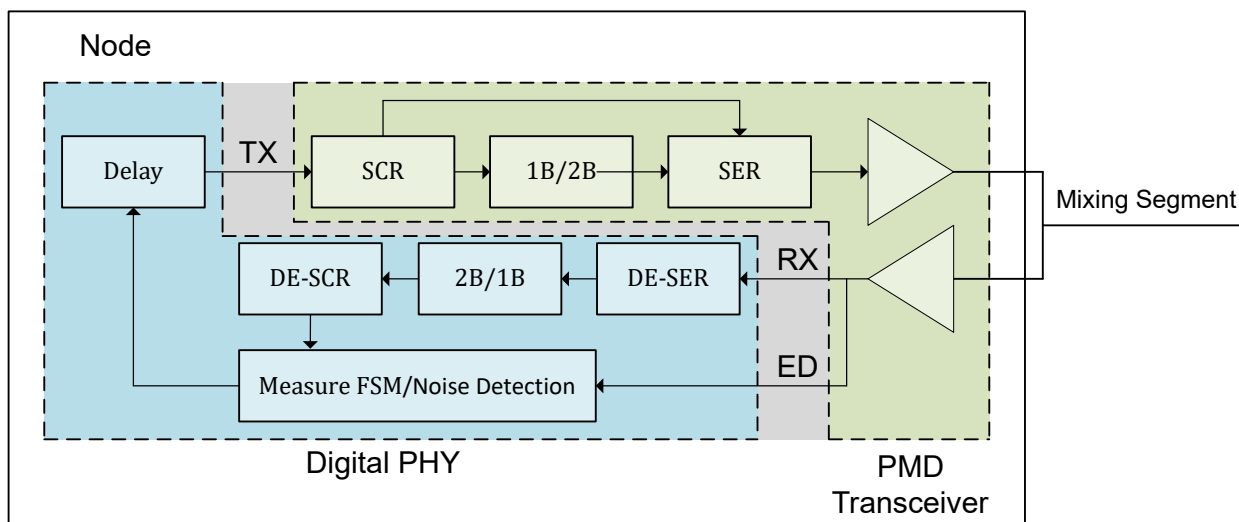
Hardware Block Diagram

Topology discovery requires that the Digital PHY and the PMD transceiver use different internal signal routing than during normal operation. Instead, they use the configuration shown in [Figure 3-1](#) where internal signals are routed through a special scrambler, encoder, and serializer used only for topology discovery.

In topology discovery, the transmit path begins with a scrambler. This randomizes the pulse polarity to assist in rejecting noise or erroneous transmissions. After the scrambler, a 1B/2B encoder is followed by a serializer; this combination turns the single pulse into a pair of pulses: either a positive pulse followed by a negative pulse, or a negative pulse followed by a positive pulse. This design prevents baseline wander on the mixing segment.

The receive path contains the complimentary deserializer, decoder and descrambler, as well as a state machine which counts the number of incoming pulses and detects errors. The output of the state machine enters a delay block, which ensures that there is a minimum total internal delay of at least 100 ns. The groups of blocks labeled Digital PHY and Transceiver may be integrated into one physical device such as Microchip's LAN8670/1/2 Revision D0 or later. Alternatively, the Digital PHY may be implemented in a device separate from the PMD transceiver if the devices are connected according to the OPEN Alliance *10BASE-T1S PMD Transceiver Interface* specification.

Figure 3-1. Block Diagram of a Node During Topology Discovery



Before the start of any measurement, the first 60 pulses are used to train the descrambler in the receiver. If synchronization between the transmitter's scrambler and the receiver's descrambler does not occur within this time or if the polarity of an incoming pulse does not match the expected polarity due to noise or error, then the measurement will stop and the status registers will indicate an error.

Distance Measurement

In topology discovery, a reference node transmits a pulse with a fixed length of 40 to 85 ns onto the mixing segment. This fixed length may vary with component number or manufacturer. The measured node detects the leading edge of the pulse and transmits a pulse in return once its internal delay has passed. The reference and measured nodes repeat this process for a configurable amount of time called the measurement *duration*. The duration can be configured between 1 to 16 ms, with a longer duration increasing the accuracy of the resulting distance calculation.

The internal delays of each node, measured as described in the next section, must be known prior to calculating the actual time delay between the devices. The distance may then be calculated using the duration, the number of pulses received and the internal delays of both nodes:

Equation 3-1. Distance measurement calculation

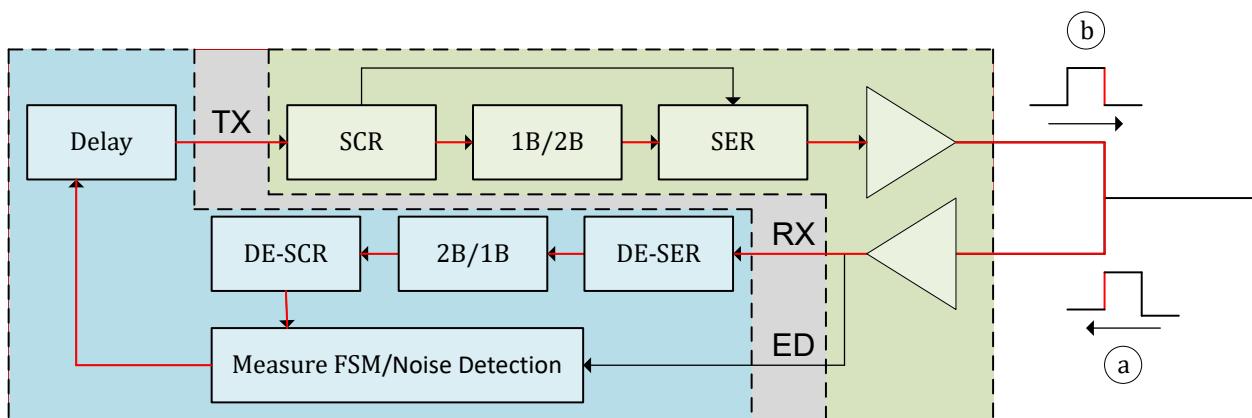
$$\text{dist} \left[\text{ns} \right] = \frac{\frac{\text{duration} \left[\text{ms} \right] * 10^6 \left[\frac{\text{ns}}{\text{ms}} \right]}{\text{pulses}} - \text{int_dly_ref} \left[\text{ns} \right] - \text{int_dly_meas} \left[\text{ns} \right]}{2}$$

Note: It is strongly recommended that the reference node is one of the end nodes of the mixing segment, as shown on the cover page. This reduces the probability of signal reflections interfering with the measurement.

Internal Delay

To calculate the time of flight of the signal using the equation above, it is necessary to know the internal delays of each of the participating nodes. Therefore, the internal delay of each participating node is measured before the distance measurement is performed. Measurement of the internal delay is more accurate than relying on a fixed value such as one from a data sheet, since measurement helps account for and eliminate uncertainties due to semiconductor manufacturing variations. Figure 3-2 shows that the internal delay is the time between the first edge of a received pulse (a) to the first edge of the pulse that is transmitted as a response (b). Within a device, this signal path is a loop and the delay around the loop can be measured at any point. To measure its own internal delay, a node will measure the time delay between transmitting consecutive pulses on the TX line. This delay includes all of the blocks shown, including the transmitting path, receiving path and the delay block.

Figure 3-2. Internal Delay Path



When measuring its own internal delay, the node transmits pulses onto the mixing segment. Instead of waiting for a response from another node, it detects the leading edge of its *own* transmitted pulse and processes it through the receiver and delay line, automatically generating another pulse.

This is repeated for the duration of the measurement while the node counts the number of pulses generated in this manner. The internal delay is calculated from the duration and the number of pulses using Equation 3-2. At the same time, other nodes on the mixing segment can detect the signals transmitted onto the segment, allowing other nodes to count the pulses as well and calculate the internal delay of the transmitting node.

Equation 3-2. Internal delay calculation

$$int_dly \left[ns \right] = \frac{duration \left[ms \right] * 10^6 \left[\frac{ns}{ms} \right]}{pulses}$$

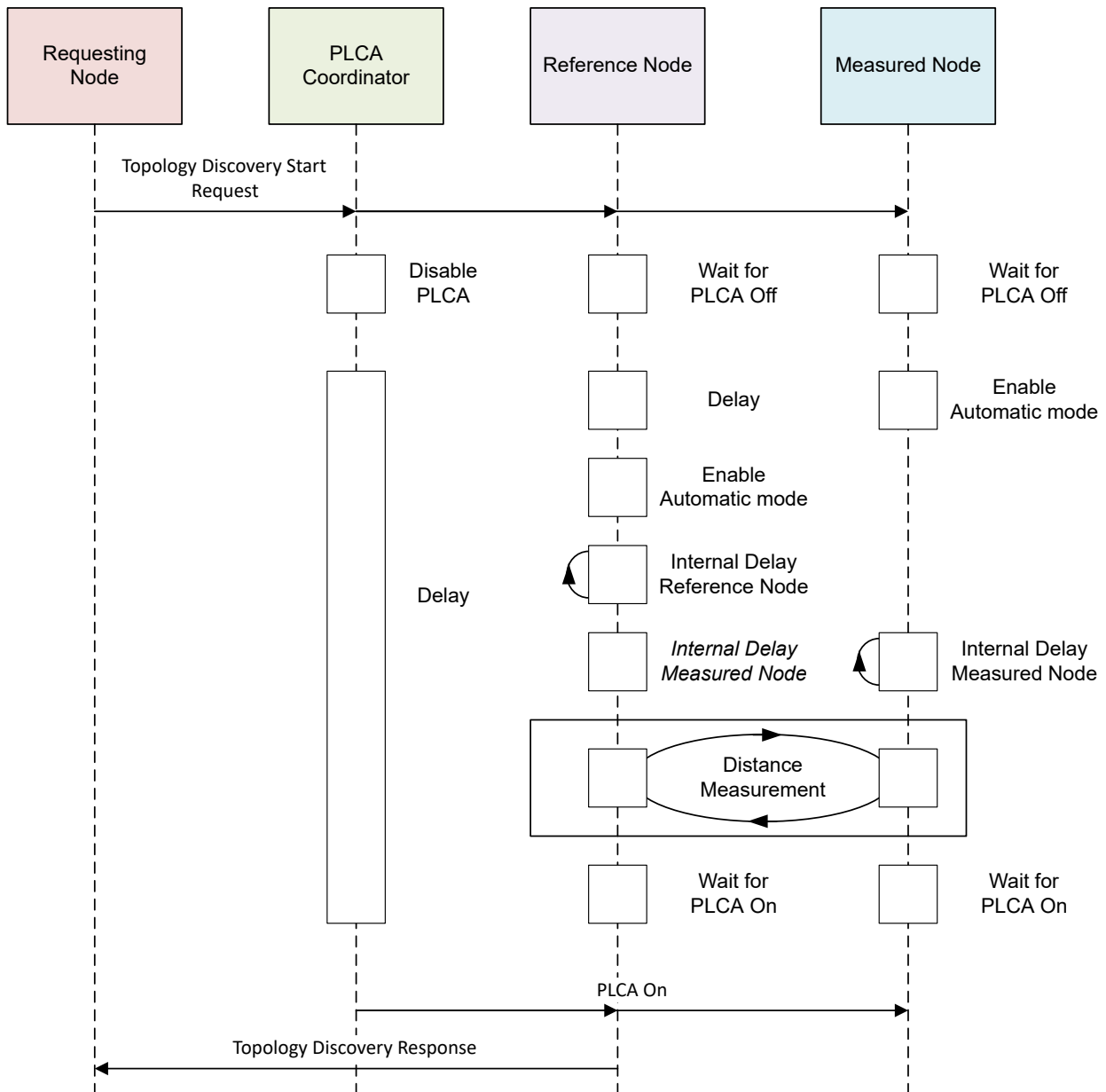
Note: During the measurement, devices react only to the leading edge of a pulse. The receive delay block ensures that there is a minimum total internal delay of at least 100 ns. Since the internal delay of a node is always greater than the maximum pulse width of 85 ns, the exact width of the pulse does not affect the delay measurement.

4. Topology Discovery Example

Figure 4-1 shows the control and process flows of the relevant nodes on the network for one complete distance measurement in automatic mode. The example below describes how to use the registers and bit-fields that are defined in the OPEN Alliance *10BASE-T1S Topology Discovery* specification to perform the measurement using the *Automatic* mode as defined in the standard.

The example proceeds as shown in Figure 4-1 and assumes that a control unit exists which configures all of the participating nodes. This control unit may be on the mixing segment or may be elsewhere in the network. It will use an appropriate protocol to identify and configure the reference and measured nodes before the measurement begins. In addition, the controller and control protocol will also need to be able to control the PLCA coordinator, which may be any node on the mixing segment.

Figure 4-1. Topology Discovery Example



Topology Discovery Start

The process begins by identifying the nodes involved in the topology discovery process and setting the measurement duration. The process shown in [Figure 4-1](#) assumes that the requesting node begins the process by sending Topology Discovery Start messages which identify the reference and measured nodes and the duration of the measurement.

The PLCA coordinator must wait until all nodes have received their configuration information, then disable PLCA to ensure that beacons are not transmitted. A simple way to disable PLCA for the coordinator is to temporarily set its node ID to 254 until the topology discovery measurement is over. After disabling PLCA, the coordinator begins an internal delay timer after which it will restart PLCA. This delay must be longer than three times the measurement duration because two internal delays and one distance measurement are performed during the procedure. An additional timing margin is recommended to account for additional delays in either the reference node or measured node before and between measurements.

All nodes not involved in the topology discovery procedure must ensure that they do not transmit during the entire measurement. In the simplest case, nodes that remain in PLCA mode cannot transmit without a beacon. However, if the beacon is disabled for more than 13 ms, a node will fallback into CSMA/CD mode. If the MAC has packets to send, a node will begin transmitting in CSMA/CD mode after this timeout. If the total time for the topology discovery process exceeds this limit, then it is necessary to use additional methods to prevent transmission. Microchip devices, including the LAN8670/1/2, provide a proprietary method of disabling the CSMA/CD fallback. The PLCA Reconciliation Sublayer Control 1 (PRSCTL1) contains a Fallback Enable (PRSCTL1.FBEN) bit, which is enabled by default. At the start of topology discovery, this bit should be cleared so that the device remains in PLCA mode regardless of the presence of a beacon.

Both the Reference Node and Measured Node need to ensure that the PLCA beacon is disabled. A simple delay timer will often be sufficient.

After PLCA Disabled - Measurement Nodes

Once the PLCA beacon has been disabled, the reference and measured nodes need to be configured by writing to the Topology Discovery Control Register (TD_CTRL) to begin operation. It is important that the measured node is configured first so that it can detect the internal delay measurement of the reference node. The measured node uses the end of the internal delay measurement of the reference node to trigger the start of its own internal delay measurement.

A node configures itself as a measured node or reference node by setting TD_CTRL.REFN to '0' or '1' respectively. The node then configures the measurement duration for both internal measurements and distance measurements by writing a value from 0 to 15 into TD_CTRL.DM_DUR[3:0], resulting in a measurement duration range between 1 to 16 ms, with a longer duration increasing the accuracy of the result. It then writes a '1' into the TD_CTRL.AUTO_START and TD_CTRL.TD_EN bits to enable topology discovery in automatic mode.

Internal Delay Measurements

The reference node first waits a small amount of time before configuring itself and activating topology discovery in automatic mode to assure that the measured node is also in topology discovery mode before beginning the procedure. The reference node first conducts its own internal delay measurement and records the result in the TD_DLY_RES_LOW and TD_DLY_RES_HIGH registers.

The measured node monitors the mixing segment to detect that the reference node has completed its internal delay measurement. Once the network has gone silent, the measured node begins its own internal delay measurement.

The reference node listens in on the internal delay measurement of the measured node and counts the number of pulses it receives. The duration of this measurement is recorded in the TD_MNDLY_DUR register. The total number of pulses counted is recorded in the TD_MNDLY_RES_LOW and TD_MNDLY_RES_HIGH registers. This allows the reference node the

calculate the internal delay of the measured node and therefore the total distance between the two nodes without needing to request this information from the measured node later.

Distance Measurement

Once the internal delay measurements are completed, the distance measurement between the measured node and reference node is started. The reference node counts how many times it receives a pulse from the measured node. This count is recorded in the TD_DIST_RES_LOW and TD_DIST_RES_HIGH registers. Once the measurement is completed, the distance between the nodes is calculated. The status of the measurements is shown in the Topology Discovery Status (TD_STAT) register. If all measurements have finished successfully, it is indicated by a '1' when reading the TD_STAT.DLYM_DONE and TD_STAT.DM_DONE bits.

Completion

At the completion of the measurement, the TD_CTRL.TD_EN bit for both the reference and measured node should be cleared to '0', ending topology discovery mode. If the proprietary PRSCTL1.FBEN bit has been cleared, it can be re-set to assure compliance with standard operation.

Once the internal delay of the coordinator is complete, it should be configured to once again send PLCA BEACONS. The reference node is now able to transmit its results to the external device in a Topology Discovery Response message.

Note: The OPEN Alliance *10BASE-T1S Topology Discovery* specification also allows for a manual operation mode. In Manual Mode, the individual steps of the process are started one at a time: internal delay measurement on the measured node, the internal delay measurement on the reference node and finally distance measurement of the reference node. These must be started sequentially by software on the control processor, which also reads back the results of each step. Manual mode may provide a quicker solution for mixing segments with many nodes, since the reference node must do its internal delay measurement only once. Manual mode and automatic mode will deliver the same results on a given mixing segment.

5. Revision History

Table 5-1.

Revision Level & Date	Section/Figure/Entry	Changes
DS00006067B (Oct-2025)	All	Made public.
DS00006067A (Aug-2025)	All	Initial revision

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